

# **Introduction to Many-Body Quantum Information**

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## 2 Hamiltonian evolution and pure-state phase

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### Main references for this section:

- Peter Wittek. *Quantum Machine Learning: What Quantum Computing Means to Data Mining*.
- Michael A. Nielsen, Isaac L. Chuang. *Quantum Computation and Quantum Information*.
- X. Chen, Z.-C. Gu, and X.-G. Wen. Phys. Rev. B 82, 155138 (2010).
- Y.-M. Ding, X. Cui, Y. Shi. Phys. Rev. D 105, 054508 (2022).

### 2.1 No-cloning theorem

For classical information, an unknown state can be copied by first performing a measurement and then preparing new systems according to the measurement outcome. A natural question is whether an analogous procedure exists for quantum states, i.e., whether an arbitrary unknown quantum state can be cloned.

To formalize this, consider an initial state  $|\psi\rangle \otimes |0\rangle$ , where  $|0\rangle$  denotes an ancillary state. A cloning operation, if it exists, is described by a unitary operator  $U$  such that

$$U(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |\psi\rangle, \quad (42)$$

for an arbitrary state  $|\psi\rangle$ . Since  $|\psi\rangle$  is unknown,  $U$  must be independent of  $|\psi\rangle$ . Similarly, for another state  $|\phi\rangle$ , one must have

$$U(|\phi\rangle \otimes |0\rangle) = |\phi\rangle \otimes |\phi\rangle. \quad (43)$$

Taking the inner product between the two cases and using the unitarity of  $U$ , we obtain

$$\begin{aligned} \langle\phi| \otimes \langle 0| |\psi\rangle \otimes |0\rangle &= \langle\phi| \otimes \langle 0| U^\dagger U |\psi\rangle \otimes |0\rangle \\ &= \langle\phi| \otimes \langle\phi| |\psi\rangle \otimes |\psi\rangle \\ &= \langle\phi|\psi\rangle^2. \end{aligned} \quad (44)$$

However, the left-hand side is simply  $\langle\phi|\psi\rangle$ , and thus we arrive at the condition

$$\langle\phi|\psi\rangle = \langle\phi|\psi\rangle^2, \quad (45)$$

which holds only if  $\langle \phi | \psi \rangle = 0$  or 1. Therefore, a universal unitary cloning operation cannot exist for arbitrary quantum states. This is the *no-cloning theorem*.

We claim that this result extends to mixed states, where it is known as the *no-broadcasting theorem*.

## 2.2 Adiabatic quantum computing

The standard setting of *adiabatic quantum computing (AQC)* is defined by an interpolating Hamiltonian

$$H(s) = sH_0 + (1 - s)H_1, \quad (46)$$

where the ground state of  $H_1$  is easy to prepare, while the ground state of  $H_0$  encodes the solution of interest.

The protocol starts from  $s = 0$ , where the system is initialized in the ground state of  $H_1$ . One then varies  $s$  monotonically from 0 to 1.

**Theorem 2.1** (*Adiabatic condition (theorem)*). *Let  $H(s)$ ,  $s \in [0, 1]$ , be a smooth family of local Hamiltonians with instantaneous eigenvalues  $E_0(s) < E_1(s) \leq \dots$ , and define the minimum spectral gap*

$$\Delta_{\min} = \min_{s \in [0,1]} (E_1(s) - E_0(s)). \quad (47)$$

*Consider the time-dependent Hamiltonian  $H(s(t))$  with  $s(0) = 0$ ,  $s(T) = 1$ . If the total evolution time satisfies*

$$T \gg \frac{\max_s \|\partial_s H(s)\|}{\Delta_{\min}^2}, \quad (48)$$

*Here  $\|\partial_s H(s)\|$  denotes the operator norm of  $\partial_s H(s)$ , i.e., the largest magnitude of its eigenvalues, which quantifies the rate of change of the Hamiltonian along the adiabatic path. Then, up to small corrections, the evolved state remains in the instantaneous ground-state manifold. In particular, the runtime scales as*

$$T \sim \Delta_{\min}^{-2}. \quad (49)$$

## 2.3 Example: 2-satisfiability (2-SAT) problem in AQC

As a concrete example, consider a 2-SAT instance with three Boolean variables  $x_1, x_2, x_3 \in \{0, 1\}$  and clauses

$$C = (\neg x_1 \vee x_2) \wedge (\neg x_2 \vee \neg x_3) \wedge (x_1 \vee x_3). \quad (50)$$

The goal is to find an assignment that satisfies all clauses<sup>1</sup>.

We map Boolean variables to qubits via

$$x_i = \frac{1 - Z_i}{2}, \quad Z_i = \pm 1, \quad (51)$$

<sup>1</sup>One can easily determine the solution for this demo problem via enumerating and obtain the two sets of solutions  $\{0, 0, 1\}$  and  $\{1, 1, 0\}$ .

where  $Z_i$  is the Pauli- $z$  operator. Each clause is encoded as a projector onto the violating subspace:

$$H^1 = \frac{1}{4}(1 + Z_1)(1 - Z_2), \quad (52)$$

$$H_2 = \frac{1}{4}(1 + Z_2)(1 + Z_3), \quad (53)$$

$$H_3 = \frac{1}{4}(1 - Z_1)(1 - Z_3), \quad (54)$$

so that each term vanishes when the clause is satisfied and is positive otherwise. The problem Hamiltonian is

$$H_0 = H_1 + H_2 + H_3 = -Z_2Z_2 + Z_2Z_3 + Z_1Z_3 + 3. \quad (55)$$

We choose the auxiliary Hamiltonian

$$H_1 = -\sum_{i=1}^3 X_i, \quad (56)$$

whose ground state is easy to prepare.

The full interpolation reads

$$H(s) = sH_0 + (1 - s)H_1. \quad (57)$$

Starting from  $s = 0$  and evolving slowly to  $s = 1$ , the system reaches the ground state of  $H_0$ , which encodes a satisfying assignment.

To implement the adiabatic evolution on a circuit-based quantum computer, we discretize the time evolution operator

$$U = \mathcal{T} \exp\left(-i \int_0^T H(s(t)) dt\right), \quad (58)$$

into small time steps  $\Delta t$ . Using the first-order Trotter–Suzuki approximation, we write

$$U \approx \prod_{k=1}^{N_t} e^{-i\Delta t s_k H_0} e^{-i\Delta t (1-s_k) H_1}, \quad (59)$$

where  $s_k = s(t_k)$ .

Since  $H_0$  consists of commuting diagonal terms in the  $Z$  basis, each factor  $e^{-i\Delta t s_k H_0}$  can be further decomposed into products of two-qubit gates (or general *Pauli-string rotation*), which is

$$e^{-iH(s(t))\Delta t} \approx e^{-i(1-s)X_1\Delta t} e^{-i(1-s)X_2\Delta t} e^{-i(1-s)X_3\Delta t} e^{isZ_1Z_2\Delta t} e^{-isZ_2Z_3\Delta t} e^{-isZ_1Z_3\Delta t}. \quad (60)$$

The transverse-field term  $H_1$  generates single-qubit rotations. In this way, the continuous adiabatic evolution is approximated by a quantum circuit composed of local gates.

## 2.4 Gapped quantum phase defined via finite-depth local unitaries

In this subsection, we introduce a modern definition of quantum phases for *gapped pure states*, based on their **structure of entanglement** rather than symmetry breaking. This viewpoint was developed by

Chen-Gu-Wen in 2010, where quantum phases are defined as *equivalence classes* under *finite-depth local unitary (LU) transformations*.

The idea is physically motivated by the quantum adiabatic theorem, which ensures that gapped ground states connected by a smooth, local, and gap-preserving path can be related by a finite-depth local unitary transformation.

**Definition 2.2.** A *finite-depth local unitary (LU) transformations* is a unitary operator of the form

$$U_{\text{LU}} = \prod_{\ell=1}^D U_{\ell}, \quad (61)$$

where  $D = \mathcal{O}(1)$  is independent of system size, and each layer  $U_{\ell}$  is a product of mutually commuting local unitaries,

$$U_{\ell} = \prod_i U_i^{(\ell)}, \quad (62)$$

with each  $U_i^{(\ell)}$  acting on a finite spatial region. Such transformations can be viewed as quantum circuits of finite depth.

**Definition 2.3.** Two gapped ground states  $|\psi_0\rangle$  and  $|\psi_1\rangle$  are said to belong to the same *quantum phase* if and only if there exists a finite-depth LU transformation such that

$$|\psi_1\rangle = U_{\text{LU}}|\psi_0\rangle. \quad (63)$$

Equivalently, two states are in the same phase if they can be connected by a continuous path of local gapped Hamiltonians without closing the energy gap.

This relation defines an *equivalence relation* on the space of gapped ground states, and a gapped quantum phase is thus identified with an *equivalence class* under finite-depth LU transformations.

This definition leads to a fundamental distinction between two classes of gapped states:

**Definition 2.4.** *Short-range entangled (SRE) states:* States that can be transformed into a product state by a finite-depth LU,

$$|\psi\rangle \xrightarrow{U_{\text{LU}}} \bigotimes_i |0\rangle_i. \quad (64)$$

All such states belong to the *trivial phase* in the absence of symmetry.

**Definition 2.5.** *Long-range entangled (LRE) states:* States that cannot be reduced to a product state by any finite-depth LU. These states possess *intrinsic topological order*.

*Topologically ordered phases* are precisely those characterized by long-range entanglement. They cannot be connected to trivial product states without either **closing the energy gap** or **applying nonlocal**

**(infinite-depth) transformations.** This provides an intrinsic, **symmetry-independent definition** of topological order.

**Remark 2.6.** We remark that the gapped phase defined here is *stable*: finite-depth LU transformations only modify short-range entanglement and therefore preserve universal, long-distance properties of the state.

In summary, quantum phases of gapped pure states are classified by their patterns of entanglement:

$$\text{Phase} \equiv \text{equivalence class under finite-depth LU.} \quad (65)$$

This definition generalizes the Landau paradigm and provides a unified description of both symmetry-breaking and topological phases.